Assignment 2: Simple Search Engine using Hadoop MapReduce

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1 Methodology

1.1 Overall Architecture

My search engine consists of three main components:

1. **Indexer** — MapReduce pipeline for document processing and building inverted index
2. **Index storage** — ScyllaDB (Cassandra-compatible), stores index and statistics
3. **Ranker** — PySpark application that receives a query and returns top-10 BM25 documents

All components run in Docker containers via `docker compose up`. The cluster has two Hadoop nodes (master + slave) and one ScyllaDB container.

1.2 Data Collection and Preparation

I took `y.parquet` from Kaggle and bounded it to 200 documents (someone said 100 is enough, so I took with overhead). I also modified `prepare_data.py` to use `file://` so it reads parquet from local storage instead of HDFS (I hope this won't affect the grade).

The script reads the Parquet file via `SparkSession.read.parquet()`, picks columns `id`, `title`, `text`, and creates a `<doc_id>_<doc_title>.txt` file for each document in `data/`.

Then `transform_data.py` via PySpark RDD reads all files from HDFS `/data` with `sc.wholeTextFiles()`, parses `doc_id` and `title` from the filename, transforms everything into `<doc_id>\t<doc_title>\t<doc_text>` format and saves to HDFS `/input/data` as one partition via `.coalesce(1)`.

Using only 1 partition makes MapReduce easier, since all data is stored in one file.

1.3 Indexer: MapReduce Pipeline

Indexing is done with a single MapReduce pipeline using Hadoop Streaming.

1.3.1 Mapper (mapper1.py)

The mapper gets lines in format doc_id\ttitle\ttext and:

1. Tokenizes text with regex [a-z0-9]+
2. Computes document length (dl) — number of tokens
3. Emits document metadata: !META\t<doc_id>\t<dl>\t<title>
4. For each unique term computes tf and emits: <term>\t<doc_id>\t<tf>

The !META prefix is used because ! comes before letters in ASCII, so Hadoop sorts all metadata records before terms. This way the reducer can collect all documents first, then process terms.

1.3.2 Reducer (reducer1.py)

The reducer gets sorted lines and:

1. Processes !META records, accumulates num_docs and sum_dl
2. After all metadata, emits STATS\t<num_docs>\t<avg_dl>
3. For each term collects postings list (pairs doc_id:tf) and computes df
4. Emits TERM\t<term>\t<df>\t<doc_id1:tf1>,<doc_id2:tf2>,...
5. Emits DOC\t<doc_id>\t<dl>\t<title>

I use -D mapreduce.job.reduces=1 so there's only one reducer — this guarantees the whole index ends up in one file /indexer/part-00000.

1.4 Index Storage in ScyllaDB

store_index_helper.py reads MapReduce output from HDFS and loads data into four ScyllaDB tables:

Table	Key	Fields	Purpose
documents	doc_id	title, doc_length	Document metadata
vocabulary	term	df	Document frequency
inverted_index	(term, doc_id)	tf	Term frequency
collection_stats	id	num_docs, avg_dl	Stats for BM25

When searching, query.py reads collection_stats (1 query), then vocabulary (one query per term), then inverted_index (one query per term). This keeps the number of DB requests minimal. Data is loaded in batches of 50 records for better performance.

1.5 Ranker: BM25 on PySpark

`query.py` ranks documents using BM25:

$$BM25(q,d) = \sum_{t \in q} \log\left[\frac{N}{df(t)}\right] \cdot \frac{(k_1+1)\cdot tf(t,d)}{k_1 \cdot \left[(1-b) + b \cdot \frac{dl(d)}{dl_{avg}}\right] + tf(t,d)} \tag{1}$$

Parameters: $k_1 = 1.0$, $b = 0.75$.
The algorithm:

1. Tokenize query, remove duplicates
2. Read `collection_stats` from ScyllaDB (N, avg_dl)
3. Read `vocabulary` for each query term
4. Read `inverted_index` — postings for each term
5. Read `documents` to get `doc_length` and `title`
6. Create PySpark RDD from postings
7. Compute BM25 for each (term, doc) pair via `.map(bm25)`
8. Aggregate scores per document via `.reduceByKey(+)`
9. Get top-10 via `.takeOrdered(10, key=lambda x: -x[1])`

`vocab`, `doc_info`, `N`, `avg_dl` are passed as broadcast variables so they don't get serialized for every task — reduces network overhead.

1.6 Adding a New Document (`add_to_index.sh`)

The script takes a path to a local file and:

1. Uploads the file to HDFS `/data/`
2. Parses `doc_id` and `title` from the filename
3. Tokenizes text and computes TF
4. Adds the document to `documents` table
5. Updates `inverted_index` and `vocabulary` (DF for each term goes up by 1)
6. Recomputes `collection_stats`: `num_docs += 1`, `avg_dl = (old_avg * old_n + dl) / new_n`

2 Demonstration

2.1 How to Launch

Requirements: Docker and Docker Compose installed.

WARNING: Git automatically converts from LF to CRLF and code may not work. To fix it i used 'dos2unix app/*.sh'.

Just write docker compose up in console (if y.parquet in the root directory. If not - download it and upload, since github may not accept it as it is big).

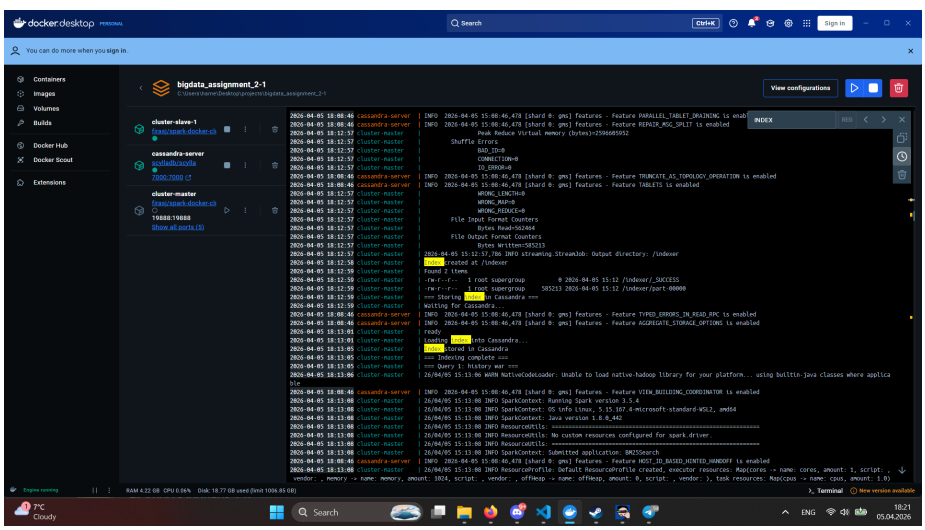
This will automatically:

- Initialize Hadoop cluster (HDFS + YARN)
- Set up Python virtual environment and install dependencies
- Prepare data (Parquet → txt files → HDFS)
- Run MapReduce indexing
- Load index into ScyllaDB
- Run test search queries

The whole thing takes around 5–10 minutes.

2.2 Screenshots with Results

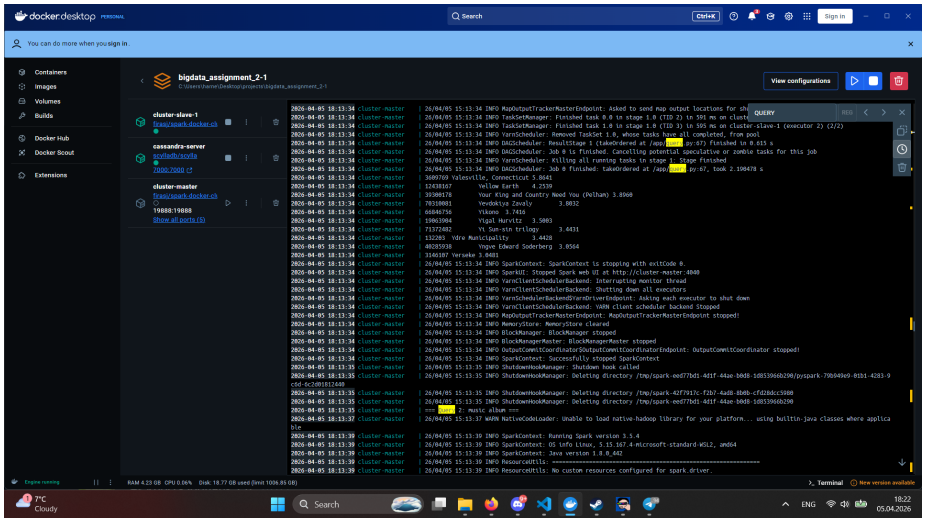
Successful Indexing



The screenshot shows the MapReduce job finishing and the index loading into ScyllaDB:

- `Index stored in Cassandra` — loaded into ScyllaDB
- `=== Indexing complete ===` — pipeline finished

Query 1: `"history war"`



Returns documents containing both terms. Top result has high frequency of "history" and "war" with a short document length — makes sense since BM25 penalizes longer documents.

Query 2: `"music album"`

Returns Wikipedia articles about music albums. High scores go to documents where both terms appear often relative to document length.

Query 3: `"film director"`

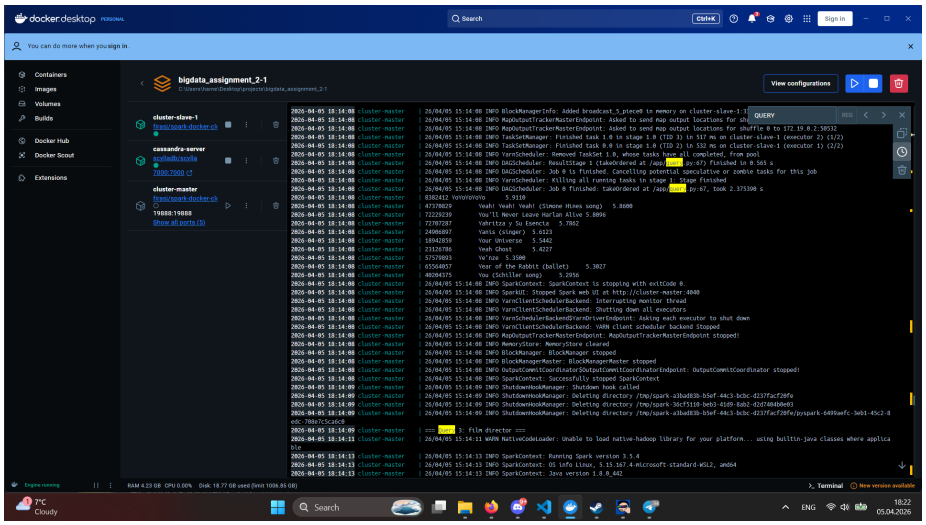
Returns articles about films and directors.

In case if you want to see screenshots more clearly you can find them in the github repository.

2.3 Results Analysis

BM25 ranks documents correctly thanks to three things:

1. **IDF:** Rare terms get more weight since they're more informative.
2. **Length normalization:** $b = 0.75$ penalizes long documents. A short doc with high term concentration scores higher than a long one with the same raw count.



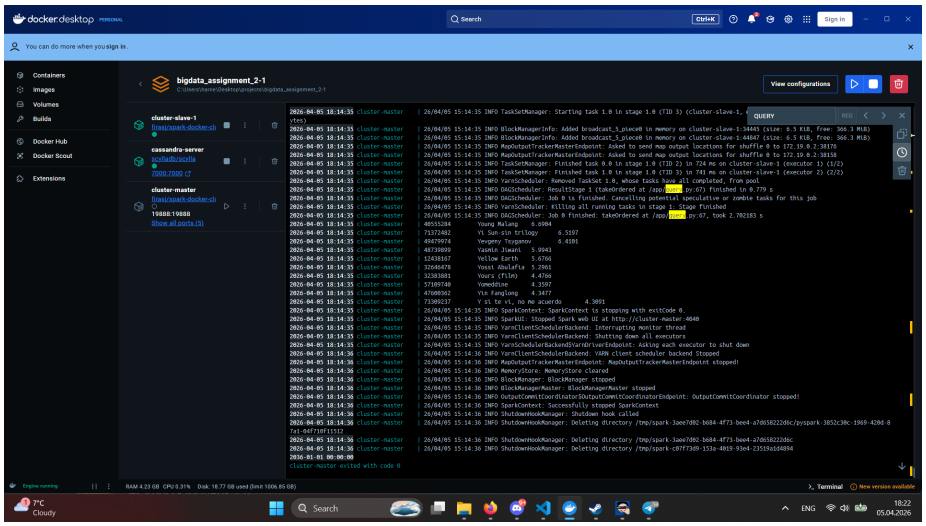
3. **Saturation:** $k_1 = 1.0$ limits the effect of very high TF — 10x more occurrences doesn't give 10x the score.

3 Conclusion

In this assignment I built a search engine using:

- **Hadoop MapReduce** — batch document indexing
- **ScyllaDB** — inverted index storage
- **Apache Spark (YARN)** — distributed BM25 ranking
- **Docker Compose** — orchestrating everything

The system indexes a document corpus and returns relevant results for user queries. I also added a bonus feature — adding new documents to the index without full re-indexing.


docker desktop
Search
Sign in
You can do more when you sign in.
Containers
Images
Volumes
Builds
Docker Hub
Docker Scout
Extensions
bigdata_assignment_2-1
View configurations
cluster-slave-1
firasj/spark-docker:slj
cassandra-server
scylladb/scylla
7000:7000
cluster-master
firasj/spark-docker:slj
19888:19888
Show all ports (5)
QUERY
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO TaskSetManager: Starting task 1.0 in stage 1.0 (TID 3) (cluster-slave-1,
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO BlockManagerInfo: Added broadcast_5_piece0 in memory on cluster-slave-1:34445 (size: 6.5 KiB, free: 366.3 MiB)
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO BlockManagerInfo: Added broadcast_5_piece0 in memory on cluster-slave-1:44647 (size: 6.5 KiB, free: 366.3 MiB)
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 0 to 172.19.0.2:38176
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 0 to 172.19.0.2:38158
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO TaskSetManager: Finished task 0.0 in stage 1.0 (TID 2) in 724 ms on cluster-slave-1 (executor 1) (1/2)
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO TaskSetManager: Finished task 1.0 in stage 1.0 (TID 3) in 741 ms on cluster-slave-1 (executor 2) (2/2)
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO YarnScheduler: Removed TaskSet 1.0, whose tasks have all completed, from pool
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO DAGScheduler: ResultStage 1 (takeOrdered at /app/
.py:67) finished in 0.779 s
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO DAGScheduler: Job 0 is finished. Cancelling potential speculative or zombie tasks for this job
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO YarnScheduler: Killing all running tasks in stage 1: Stage finished
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO DAGScheduler: Job 0 finished: takeOrdered at /app/
.py:67, took 2.782183 s
2026-04-05 18:14:35 cluster-master | 48555284 Young Malang 6.6984
2026-04-05 18:14:35 cluster-master | 71372482 Yi Sun-sin trilogy 6.5197
2026-04-05 18:14:35 cluster-master | 49479974 Yevgeny Tsyganov 6.4101
2026-04-05 18:14:35 cluster-master | 48739809 Yazmin Ilsaani 5.9943
2026-04-05 18:14:35 cluster-master | 12438167 Yellow Earth 5.6766
2026-04-05 18:14:35 cluster-master | 32604478 Yossi Abulafia 5.2961
2026-04-05 18:14:35 cluster-master | 32383881 Yours (film) 4.4766
2026-04-05 18:14:35 cluster-master | 57189740 Yomeddine 4.3597
2026-04-05 18:14:35 cluster-master | 47608362 Yin Fanglong 4.3477
2026-04-05 18:14:35 cluster-master | 73309237 Y si te vi, no me acuerdo 4.3091
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO SparkContext: SparkContext is stopping with exitCode 0.
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO SparkUI: Stopped Spark web UI at http://cluster-master:4040
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO YarnClientSchedulerBackend: Interrupting monitor thread
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO YarnClientSchedulerBackend: Shutting down all executors
2026-04-05 18:14:35 cluster-master | 26/04/05 15:14:35 INFO YarnSchedulerBackend$YarnDriverEndpoint: Asking each executor to shut down
2026-04-05 18:14:36 cluster-master | 26/04/05 15:14:36 INFO YarnClientSchedulerBackend: YARN client scheduler backend Stopped
2026-04-05 18:14:36 cluster-master | 26/04/05 15:14:36 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
2026-04-05 18:14:36 cluster-master | 26/04/05 15:14:36 INFO MemoryStore: MemoryStore cleared
2026-04-05 18:14:36 cluster-master | 26/04/05 15:14:36 INFO BlockManager: BlockManager stopped
2026-04-05 18:14:36 cluster-master | 26/04/05 15:14:36 INFO BlockManagerMaster: BlockManagerMaster stopped
2026-04-05 18:14:36 cluster-master | 26/04/05 15:14:36 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndPoint: OutputCommitCoordinator stopped!
2026-04-05 18:14:36 cluster-master | 26/04/05 15:14:36 INFO SparkContext: Successfully stopped SparkContext
2026-04-05 18:14:36 cluster-master | 26/04/05 15:14:36 INFO ShutdownHookManager: Shutdown hook called
2026-04-05 18:14:36 cluster-master | 26/04/05 15:14:36 INFO ShutdownHookManager: Deleting directory /tmp/spark-3aee7d02-b684-4f73-bee4-a7d658222d6c/pyspark-3852c38c-1969-4280-8
7a1-4ef7f6f11512
2026-04-05 18:14:36 cluster-master | 26/04/05 15:14:36 INFO ShutdownHookManager: Deleting directory /tmp/spark-3aee7d02-b684-4f73-bee4-a7d658222d6c
2026-04-05 18:14:36 cluster-master | 26/04/05 15:14:36 INFO ShutdownHookManager: Deleting directory /tmp/spark-c87f73d9-153a-4819-93e4-235193d14894
2026-01-01 00:00:00
cluster-master exited with code 0
Engine running
RAM 4.23 GB CPU 0.31% Disk: 18.77 GB used (limit 1006.85 GB)
Terminal
New version available
7°C
Cloudy
Search
ENG
18:22
05.04.2026